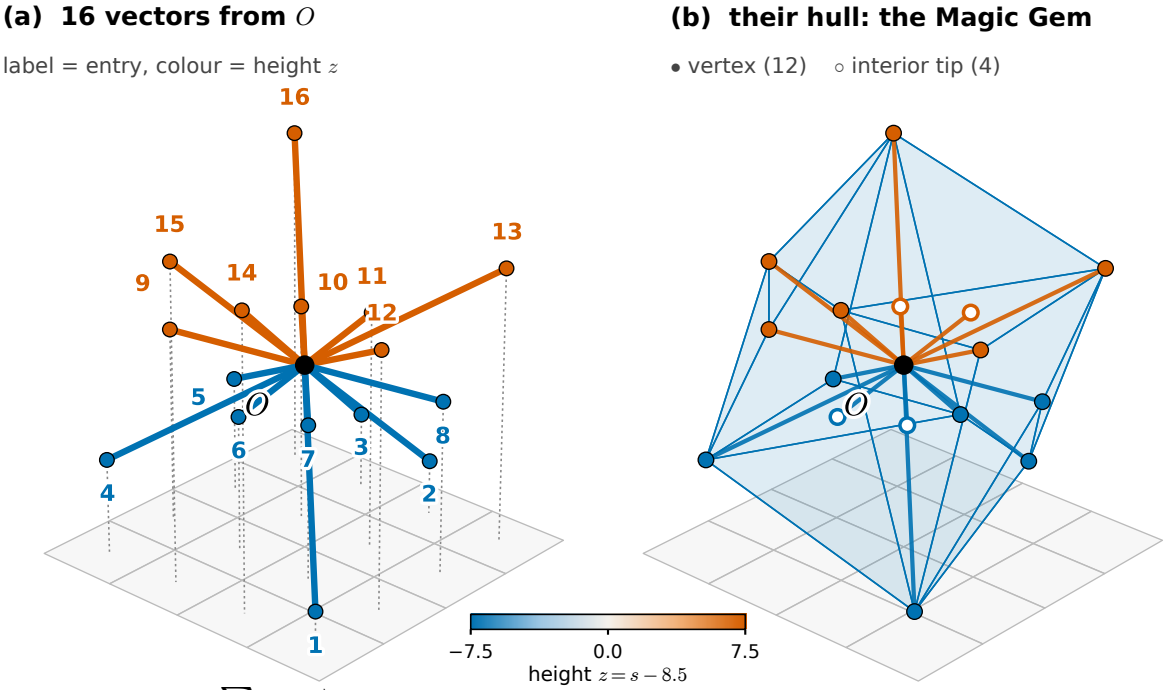

$\sum \vec{p}_{ij} = \vec{0}$ for every arrangement: O is the centroid of the 16 tips